

Zabalo Report

Morgan Manning

8/15/2023

Introduction

Mammal population dynamics are closely linked to human activity (Lee 1987, Ritzel and Gallo 2020). This tie of human culture to the environment is known as “biocultural heritage” (Davidson-Hunt et al. 2012, Gavin et al. 2015). Biocultural heritage includes the practices and knowledge of indigenous communities and positively affects local wildlife populations. Many of these indigenous communities have faced a loss of biocultural heritage (Davidson-Hunt et al. 2012, Gavin et al. 2015, Rotherham 2015). With this loss of biocultural heritage, indigenous communities interact less directly with the environment, but interactions tend to be more extractive than in communities with a stronger link to biocultural heritage (Peres 2001, Gavin et al. 2015, Rotherham 2015, de Araujo Lima Constantino 2016). One such example of this is hunting. Indigenous communities with strong ties to their biocultural heritage rely on daily subsistence hunting whereas communities with detachment from biocultural heritage rely on daily or weekly markets for food. Studies have shown that hunting increases with a community’s access to a market, where individuals are able to sell excess meat (de la Montaña et al. 2015, de Araujo Lima Constantino 2016). Conversely, communities that rely on subsistence hunting have lower harvest rates. This preliminary report will focus on Zabalo, one of the four indigenous communities in the Ecuadorian Amazon that will be included in our research on mammal occupancy across the gradient of detachment from biocultural heritage. Zabalo represents the community with the strongest ties to the environment, with members relying on daily subsistence hunting.

Methodology

We used single-species, single-season occupancy models within the *unmarked* package to explore the occupancy of three common species in the study area: the collared peccary, the lowland paca, and the brown brocket. Habitat type (lowland, swamp, or highland), effort (the number of days the camera at that site was operational), and whether the camera site was located on a trail were included as covariates of detection. Binary hunting presence, habitat type (lowland, swamp, or highland), hunting intensity, and the distances to Zabalo, rivers, and trails were considered as covariates to model occupancy. Hunting intensity was calculated for each camera trap site like so:

$$Hunting\ Intensity_j = \sum_{i=1}^{777} e^{-(distance_{ij})}$$

where $distance_{ij}$ is the distance between hunting occasion i and camera trap site j .

All continuous covariates were normalized. All combinations of occupancy and detection covariates ($n = 512$) were modeled for each species using the *unmarked* package with the number of iterations increased to 10,000 and providing a starting value of 0 for all parameters (Fiske and Chandler 2011). Models with convergence problems and with standard errors larger than the absolute value of the intercept estimate were removed.

Results

Collared peccary

Best Model (based on AIC)

```
##
## Call:
## occu(formula = formula(unmarkedModels[1, 1]), data = peccaryUF0,
##       starts = c(rep(0, length(test@opt$par))), control = 10000)
##
## Occupancy:
##               Estimate      SE      z P(>|z|)
## (Intercept)      2.05   1.92 1.070 0.2846
## Community        7.01   3.42 2.048 0.0405
## HabitatSwamp     8.72  70.19 0.124 0.9011
## HabitatUpland    2.59   2.15 1.209 0.2265
## HuntingIntensity  5.17   2.90 1.781 0.0748
##
## Detection:
##      Estimate      SE      z P(>|z|)
##    -0.0183 0.226 -0.0811 0.935
##
## AIC: 138.2431
```

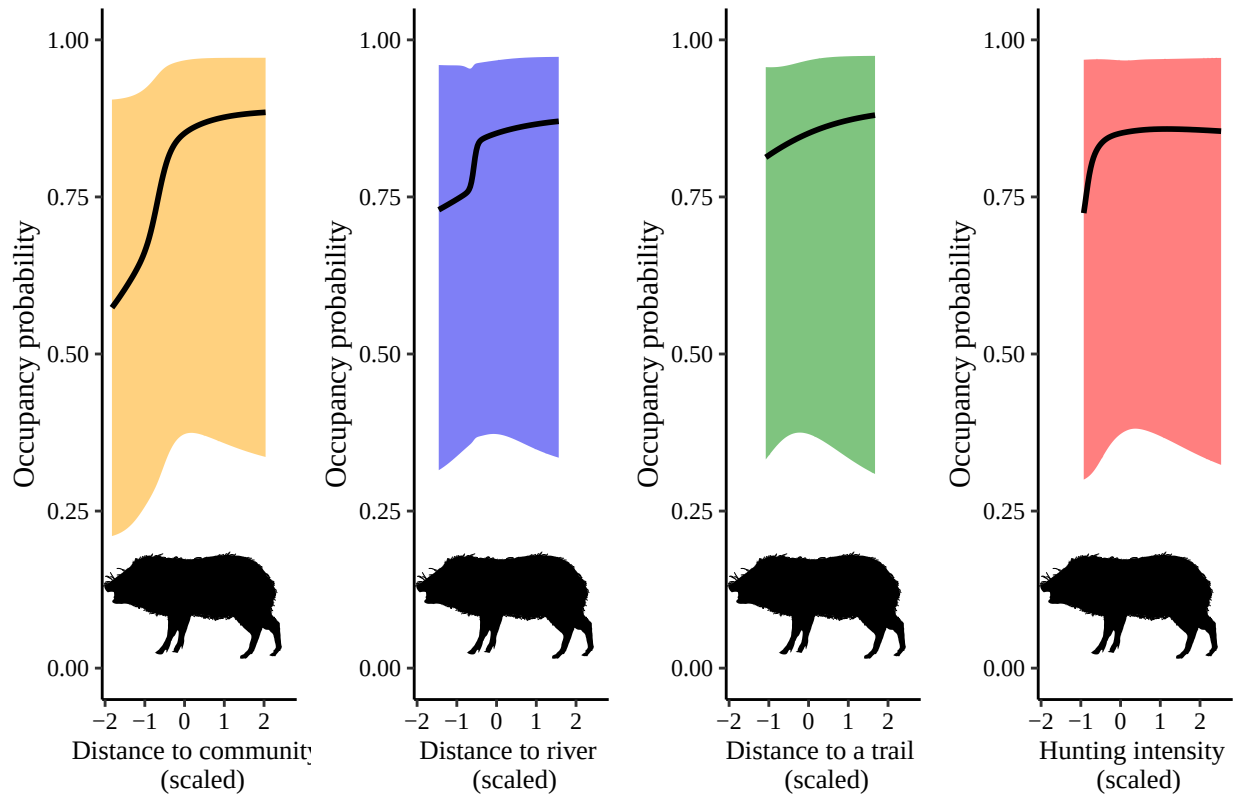
Top Models (all models within 2 of the lowest AIC)

##	formula	AIC	AICwt
## 104	~1 ~ Community + Habitat + HuntingIntensity	138.2431	0.06372230
## 105	~1 ~ River + Hunting + HuntingIntensity	138.3045	0.06179774
## 78	~Effort ~ Community + Habitat + HuntingIntensity	138.8618	0.04676803
## 91	~1 ~ Community	139.3395	0.03683093
## 92	~1 ~ River	139.7046	0.03068589
## 84	~Effort ~ 1	139.7247	0.03037945
## 8	~Effort ~ River	140.0656	0.02561843
## 79	~OnTrail ~ Community + Habitat + HuntingIntensity	140.2249	0.02365660
## 94	~1 ~ Trail.Distance	140.2293	0.02360439
## 1	~Effort ~ Community	140.2343	0.02354558

Prediction Plots

Occupancy probability is plotted across each normalized covariate by holding all other continuous covariates at their means or at the mode for factor covariates. All models (after the removal of models with convergence issues and with standard errors larger than the intercept estimate) were weighted and averaged to predict occupancy.

Collared peccary



Lowland paca

Best Model (based on AIC)

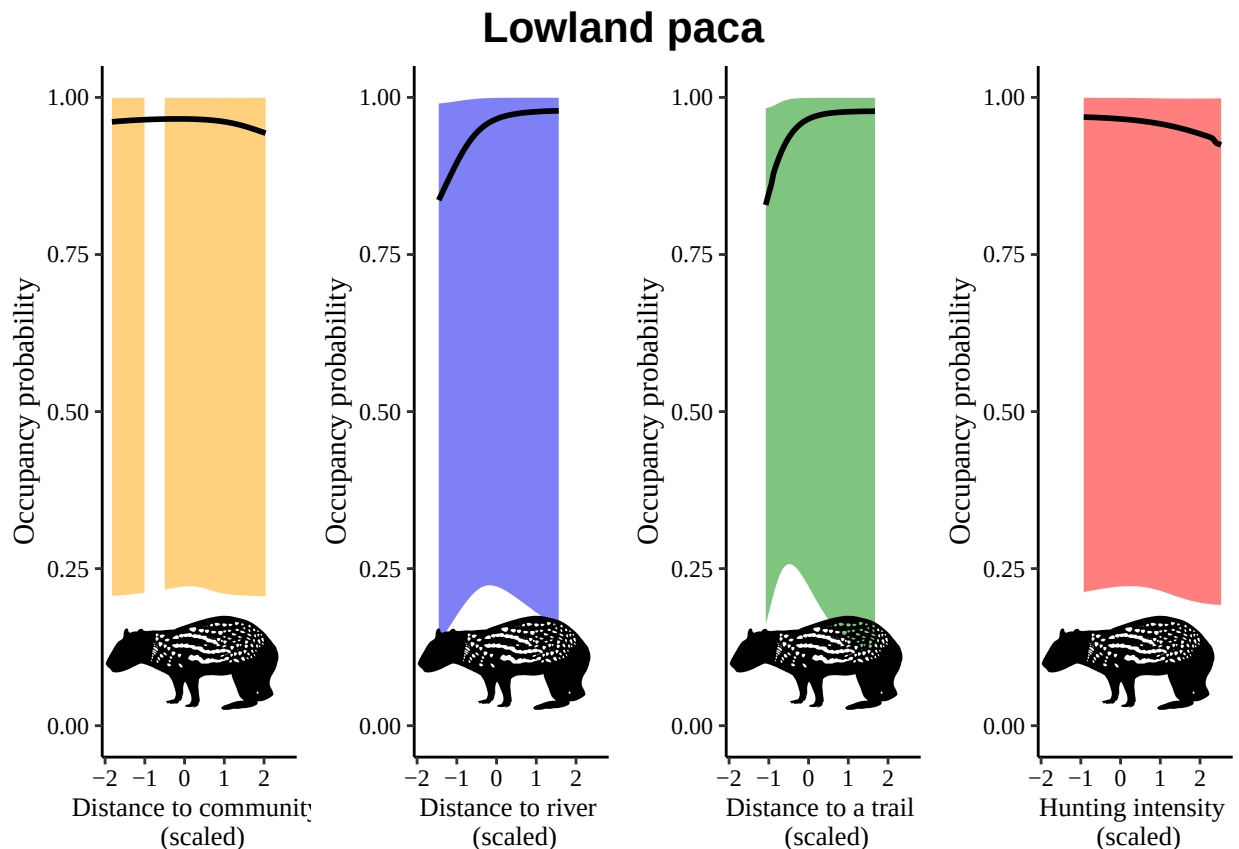
```
##
## Call:
## occu(formula = formula(unmarkedModels[1, 1]), data = pacaUFO,
##       starts = c(rep(0, length(test@opt$par))), control = 10000)
##
## Occupancy:
##               Estimate    SE      z P(>|z|)
## (Intercept)      3.45 2.28 1.52  0.129
## Trail.Distance    2.79 2.43 1.15  0.250
##
## Detection:
##               Estimate    SE      z P(>|z|)
## (Intercept)    -2.792 0.307 -9.08 1.07e-19
## Effort          0.724 0.194  3.74 1.88e-04
## HabitatSwamp    0.881 0.435  2.02 4.30e-02
## HabitatUpland  -0.484 0.349 -1.39 1.65e-01
##
## AIC: 448.8441
```

Top Models (all models within 2 of the lowest AIC)

##	formula	AIC	AICwt
## 19	~Effort + Habitat ~ Trail.Distance	448.8441	0.13484757
## 133	~Effort + Habitat ~ 1	448.9915	0.12526448
## 7	~Effort + Habitat ~ River	449.3817	0.10306246
## 53	~Effort + Habitat ~ River + Trail.Distance	449.9620	0.07710434
## 25	~Effort + Habitat ~ HuntingIntensity	450.3556	0.06333173
## 37	~Effort + Habitat ~ Community + Trail.Distance	450.5419	0.05769887
## 21	~Effort + Habitat + OnTrail ~ Trail.Distance	450.5802	0.05660438

Prediction Plots

Occupancy probability is plotted across each normalized covariate by holding all other continuous covariates at their means or at the mode for factor covariates. All models (after the removal of models with convergence issues and with standard errors larger than the intercept estimate) were weighted and averaged to predict occupancy.



Brown brocket

Best Model (based on AIC)

```
##
## Call:
## occu(formula = formula(unmarkedModels[1, 1]), data = brocketUF0,
##       starts = c(rep(0, length(test@opt$par))), control = 10000)
```

```
##
## Occupancy:
##      Estimate      SE      z P(>|z|)
## (Intercept)      1.32 0.785 1.68 0.0939
## Community        1.37 0.729 1.88 0.0607
##
## Detection:
##      Estimate      SE      z P(>|z|)
## (Intercept)     -0.654 0.744 -0.880 0.3789
## Effort           1.472 0.697 2.113 0.0346
## HabitatSwamp     -9.630 86.032 -0.112 0.9109
## HabitatUpland    2.278 1.102 2.068 0.0387
##
## AIC: 59.75012
```

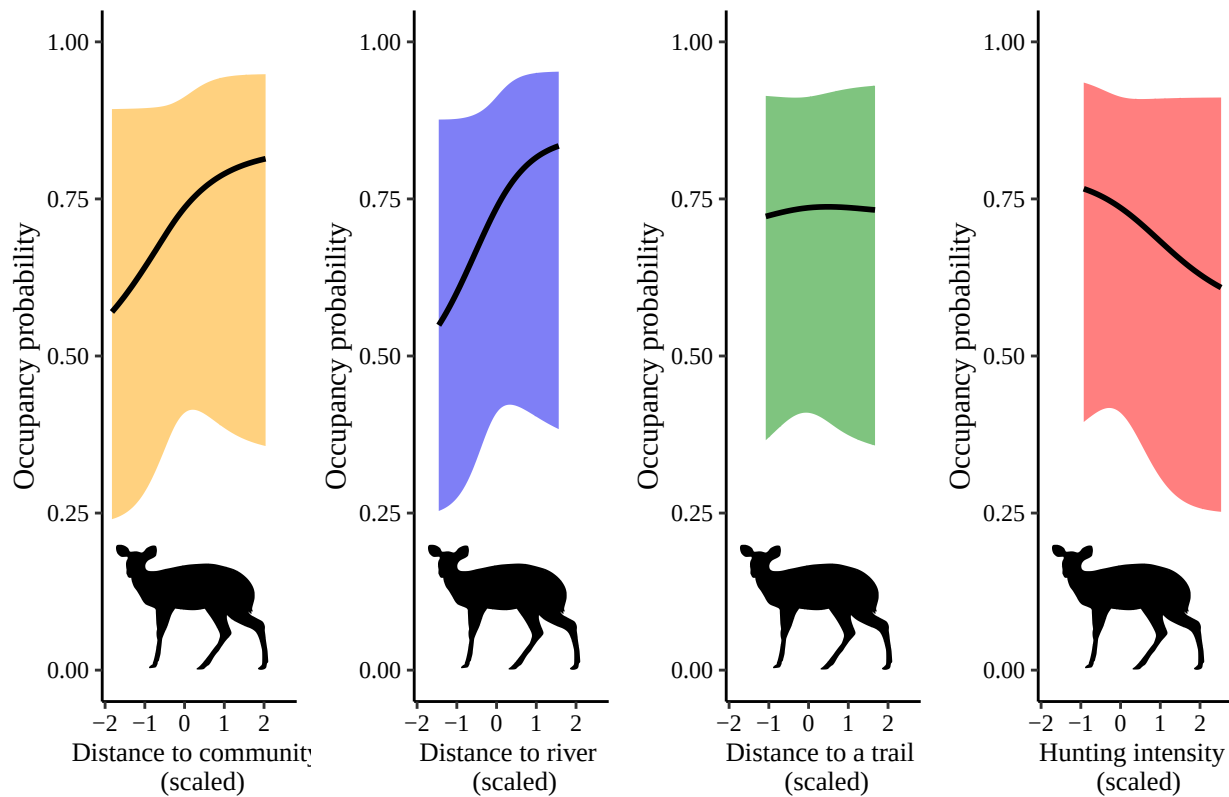
Top Models (all models within 2 of the lowest AIC)

##	formula	AIC	AICwt
## 3	~Effort + Habitat ~ Community	59.75012	0.08973342
## 26	~Effort + Habitat ~ HuntingIntensity	60.66434	0.05681120
## 6	~Habitat ~ River	61.03108	0.04729289
## 32	~Effort + Habitat ~ Community + River	61.41579	0.03901737
## 48	~Effort + Habitat ~ River + HuntingIntensity	61.51054	0.03721198
## 9	~Effort + Habitat + OnTrail ~ River	61.51536	0.03712251
## 41	~Effort + Habitat ~ Community + HuntingIntensity	61.63705	0.03493102
## 5	~Effort + Habitat + OnTrail ~ Community	61.74056	0.03316926

Prediction Plots

Occupancy probability is plotted across each normalized covariate by holding all other continuous covariates at their means or at the mode for factor covariates. All models (after the removal of models with convergence issues and with standard errors larger than the intercept estimate) were weighted and averaged to predict occupancy.

Red brocket



Bibliography

1. de Araujo Lima Constantino, P. 2016. Deforestation and hunting effects on wildlife across Amazonian indigenous lands. *Ecology and Society* 21. <https://www.jstor.org/stable/26270361>. Accessed 15 Apr 2023.
2. Davidson-Hunt, I. J., K. L. Turner, A. T. P. Mead, J. Cabrera-Lopez, R. Bolton, C. J. Idrobo, I. Miretski, A. Morrison, and J. P. Robson. 2012. Biocultural Design: A New Conceptual Framework for Sustainable Development in Rural Indigenous and Local Communities. *S.A.P.I.E.N.S. Surveys and Perspectives Integrating Environment and Society*. <https://journals.openedition.org/sapiens/1382>. Accessed 3 Aug 2023.
3. Fiske, I., and R. Chandler. 2011. unmarked: An R Package for Fitting Hierarchical Models of Wildlife Occurrence and Abundance. *Journal of Statistical Software* 43:1–23.
4. Gavin, M. C., J. McCarter, A. Mead, F. Berkes, J. R. Stepp, D. Peterson, and R. Tang. 2015. Defining biocultural approaches to conservation. *Trends in Ecology & Evolution* 30:140–145.
5. Lee, R. D. 1987. Population Dynamics of Humans and Other Animals. *Demography* 24:443–465.
6. de la Montaña, E., R. del Pilar Moreno-Sánchez, J. H. Maldonado, and D. M. Griffith. 2015. Predicting hunter behavior of indigenous communities in the Ecuadorian Amazon: insights from a household production model. *Ecology and Society* 20. <https://www.jstor.org/stable/26270294>. Accessed 15 Apr 2023.
7. Peres, C. A. 2001. Synergistic Effects of Subsistence Hunting and Habitat Fragmentation on Amazonian Forest Vertebrates. *Conservation Biology* 15:1490–1505.
8. Ritzel, K., and T. Gallo. 2020. Behavior Change in Urban Mammals: A Systematic Review. *Frontiers in Ecology and Evolution* 8. <https://www.frontiersin.org/articles/10.3389/fevo.2020.576665>. Accessed 3 Aug 2023.

9. Rotherham, I. D. 2015. Bio-cultural heritage and biodiversity: emerging paradigms in conservation and planning. *Biodiversity and Conservation* 24:3405–3429.